1.Newspaper subscription java

```
abstract class NewspaperSubscription {
    protected String name;
    protected String address;
    protected double rate;

    public NewspaperSubscription(String name) {
        this.name = name;
    }

    public String getName() {
        return name;
    }

    public String getAddress() {
        return address;
    }

    public double getRate() {
        return rate;
    }

    public abstract void setAddress(String address);
}
```

Physical newspaper subscription java

```
class PhysicalNewspaperSubscription extends NewspaperSubscription {

    public PhysicalNewspaperSubscription(String name) {
        super(name);
```

```
    }

    @Override
    public void setAddress(String address) {
        if (address.matches(".*\\d.*")) { // Check if address contains at least one digit
            this.address = address;
            this.rate = 15.0;
        } else {
            this.address = "Invalid Address";
            this.rate = 0.0;
            System.out.println("Error: Physical address must contain at least one digit.");
        }
    }
}
```

Online newspaper subscription java

```
class OnlineNewspaperSubscription extends NewspaperSubscription {

    public OnlineNewspaperSubscription(String name) {
        super(name);
    }

    @Override
    public void setAddress(String address) {
        if (address.contains("@")) { // Check if address contains '@'
            this.address = address;
            this.rate = 9.0;
        } else {
            this.address = "Invalid Address";
            this.rate = 0.0;
            System.out.println("Error: Online address must contain '@'.");
```

```
    }
  }
}
```

Demo Subscription java

```
public class DemoSubscriptions {
  public static void main(String[] args) {
    NewspaperSubscription physicalSub = new PhysicalNewspaperSubscription("John Doe");
    physicalSub.setAddress("123 Main St");

    NewspaperSubscription invalidPhysicalSub = new PhysicalNewspaperSubscription("Jane Smith");
    invalidPhysicalSub.setAddress("Main Street"); // Invalid address

    NewspaperSubscription onlineSub = new OnlineNewspaperSubscription("Alice Brown");
    onlineSub.setAddress("alice@example.com");

    NewspaperSubscription invalidOnlineSub = new OnlineNewspaperSubscription("Bob Green");
    invalidOnlineSub.setAddress("bobexample.com"); // Invalid address

    // Displaying subscription details
    System.out.println("\nSubscriptions:");
    System.out.println(physicalSub.getName() + " - Address: " + physicalSub.getAddress() + ", Rate: $" + physicalSub.getRate());
    System.out.println(invalidPhysicalSub.getName() + " - Address: " + invalidPhysicalSub.getAddress() + ", Rate: $" + invalidPhysicalSub.getRate());
    System.out.println(onlineSub.getName() + " - Address: " + onlineSub.getAddress() + ", Rate: $" + onlineSub.getRate());
    System.out.println(invalidOnlineSub.getName() + " - Address: " + invalidOnlineSub.getAddress() + ", Rate: $" + invalidOnlineSub.getRate());
  }
```

```
}
```

2.a) Turner.java

```
      interface Turner {
  void turn();
}
```

Leaf.java

```
class Leaf implements Turner {
  @Override
  public void turn() {
    System.out.println("Changing colors.");
  }
}
```

Page.java

```
class Page implements Turner {
  @Override
  public void turn() {
    System.out.println("Going to the next page.");
  }
}
```

Pancake.java

```
class Pancake implements Turner {
  @Override
  public void turn() {
    System.out.println("Flipping.");
  }
}
```

Demo Turners.java

```
public class DemoTurners {
   public static void main(String[] args) {
      Turner leaf = new Leaf();
      Turner page = new Page();
      Turner pancake = new Pancake();

      System.out.println("Turner actions:");
      leaf.turn();
      page.turn();
      pancake.turn();
   }

}
```

2.b) Wheel.java

```
class Wheel implements Turner {
   @Override
   public void turn() {
      System.out.println("Rotating forward.");
   }
}
```

Fan.java

```
class Fan implements Turner {
   @Override
   public void turn() {
      System.out.println("Spinning rapidly.");
   }
}
```

Demo Turners2.java

```
public class DemoTurners2 {
   public static void main(String[] args) {
      Turner leaf = new Leaf();
      Turner page = new Page();
      Turner pancake = new Pancake();
      Turner wheel = new Wheel();
      Turner fan = new Fan();

      System.out.println("Turner actions:");
      leaf.turn();
      page.turn();
      pancake.turn();
      wheel.turn();
      fan.turn();
   }
}
```

OUTPUT EXAMPLES

Demo Subscriptions.java

```
Error: Physical address must contain at least one digit.
Error: Online address must contain '@'.

Subscriptions:
John Doe - Address: 123 Main St, Rate: $15.0
Jane Smith - Address: Invalid Address, Rate: $0.0
Alice Brown - Address: alice@example.com, Rate: $9.0
Bob Green - Address: Invalid Address, Rate: $0.0
```

Demo Turners.java

```
Turner actions:
```

Changing colors.

Going to the next page.

Flipping.

Demo Turners2.java

Turner actions:

Changing colors.

Going to the next page.

Flipping.

Rotating forward.

Spinning rapidly.